

STOPWORD IMPACT

ANALYSIS ON NLP TASKS



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5 Strategies

3 Classifiers

15 Experiments

Reuters-21578

01 — BACKGROUND & MOTIVATION

Stopword removal is a universal NLP preprocessing step — yet its quantitative impact across classifiers and strategies is rarely studied in a controlled, multi-model setting.

This project: 15-experiment benchmark on Reuters-21578 → 5 Stopword Strategies × 3 Classifiers × 4 Metrics



RQ1

Does stopwords removal improve, degrade, or leave unchanged classification performance?



RQ2

Which strategy best balances performance with feature-space economy?



RQ3

Can a data-driven adaptive strategy outperform static lists?



RQ4

What computational efficiency gains come from aggressive vocabulary pruning?

02 — DATASET: REUTERS-21578

10,324

Filtered Articles

57 topic categories

38,858

Unique Tokens

Full vocabulary

36.74%

Stopword Ratio

Average per doc

136

Mean Doc Length

Median: 90 words

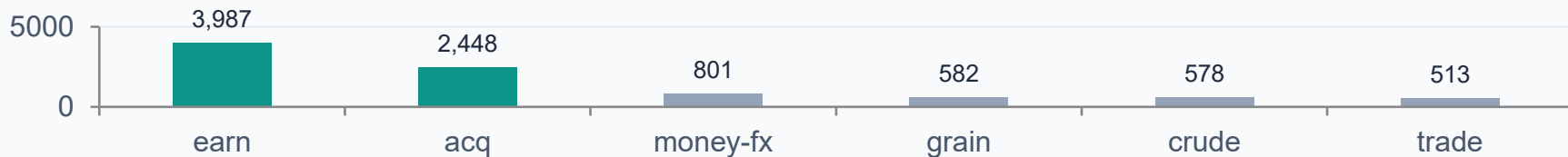
Corpus Characteristics

- **Highly skewed distribution:** earn (3,987 docs) and acq (2,448 docs) dominate
- Document lengths range from **2 to 1,669 words**
- TF-IDF vocab without removal: **14,794 features**

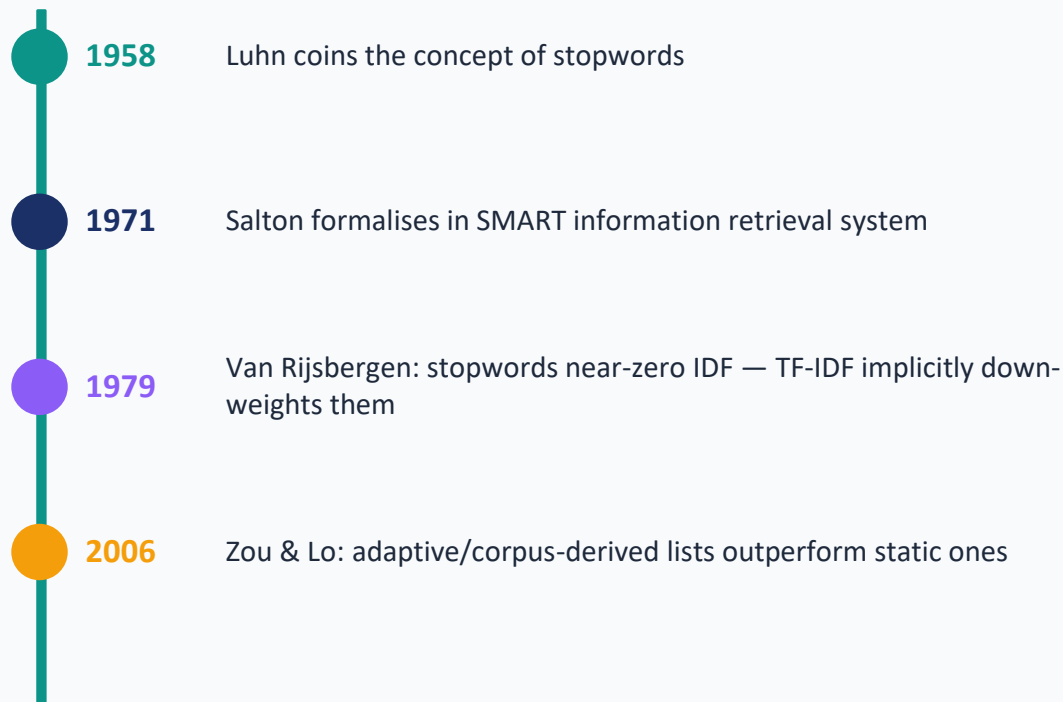
Distribution Insight

- Corpus follows **Zipf's law** — confirming stopwords removal assumptions
- **Weighted F1-score** used as primary metric due to class imbalance
- 21,578 original articles across 22 SGML files (Reuters, 1987)

Top 6 Categories by Document Count



03 — SURVEY & LITERATURE REVIEW



⚡ Research Gap

No prior study benchmarks 5 strategies × 3 classifiers on Reuters-21578 with full efficiency quantification.

Classifier Sensitivity to Stopwords

Naive Bayes

HIGH

High-freq tokens inflate class probabilities equally
[McCallum & Nigam, 1998]

SVM

LOW

Margin maximisation implicitly down-weights noise features
[Joachims, 1998]

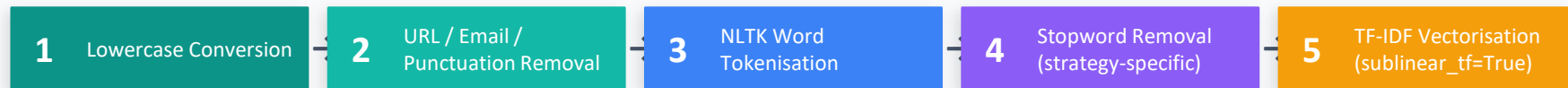
Logistic Regression

MED

Sensitivity lies between NB and SVM extremes
[Uysal & Gunal, 2014]

04 — IMPLEMENTATION

Preprocessing Pipeline



Experimental Protocol

- **Split:** 80/20 stratified, random_state=42
- 5 strategies × 3 classifiers = **15 configurations**
- Identical conditions across all experiments

Codebase

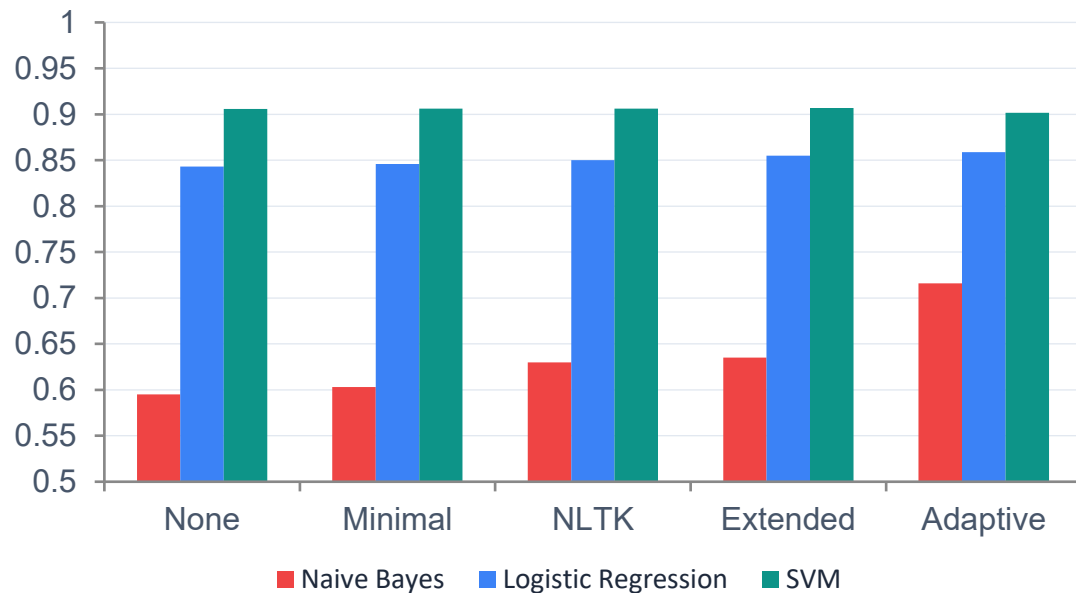
- 5 Jupyter Notebooks
- 9 src/ modules
- **106/106 unit tests passing ✓**

The 5 Stopword Strategies

none	minimal	NLTK	extended	adaptive
Full vocab baseline 14,794 features	10 core function words 14,786 features	179 standard stopwords 14,673 features	NLTK + Reuters tokens (mln, pct, dlrs...)	Corpus-derived $\theta=0.0005 \rightarrow 1,836$ features

05 — EXPERIMENTAL RESULTS

F1-Score by Classifier & Stopword Strategy



Feature Count by Strategy



Best Static

F1 = 0.9070

Extended + SVM

Best NB Gain

+20.3%

Adaptive strategy

LR Speedup

3.8×

73.9% faster training

Max Compression

87.59%

14,794 → 1,836 features

06 — ANALYSIS: PER-CLASSIFIER DEEP DIVE

Naive Bayes

MOST SENSITIVE

0.7160

- none (0.595) → minimal (0.603) → NLTK (0.630) → extended (0.635) → adaptive (0.716)
- Monotonic improvement — 20.3% relative F1 gain: largest in the entire study
- MultinomialNB estimates $P(w|c)$ per token — low-IDF tokens produce near-equal estimates across classes, adding cumulative noise

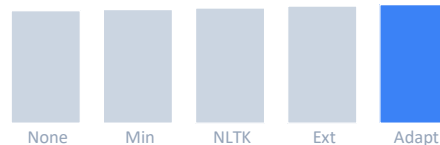


Logistic Regression

MODERATE

0.8588

- none (0.843) → extended (0.855) → adaptive (0.859), total gain +1.9%
- Training time drops 73.9% (11.35s → 2.96s) under adaptive strategy — 3.8× speedup
- LR regularisation partially controls the impact of noisy features

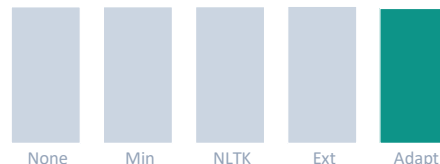


SVM

MOST ROBUST

0.9015

- none (0.906) → extended (0.907) → adaptive (0.902) — only 0.55 pp total spread
- LinearSVC margin maximisation implicitly assigns near-zero weights to non-discriminative features
- Explicit stopwords removal largely redundant — choose strategy on efficiency grounds



07 — INNOVATION: ADAPTIVE TF-IDF STRATEGY

⚡ Problem: NLTK removes only 121 tokens (<1% of vocab). Reuters-specific noise words (mln, pct, reuter, dlrs) appear in no standard list.

1

Fit TF-IDF on training data only

No test leakage — vocabulary built exclusively from training split

2

Compute mean TF-IDF per token

$\bar{s}(w) = (1/n) \sum M_{\{i,w\}}$ across all n documents in training set

3

Apply threshold $\theta = 0.0005$

Remove all tokens where $\bar{s}(w) < \theta$ — identifies 12,958 low-information tokens

Results vs Baseline:

Model	Baseline F1	Adaptive F1	Change	Speed up
Naive Bayes	0.5951	0.7160	+20.3% ↑	1.8×
Log. Regression	0.8430	0.8588	+1.9% ↑	3.8×
SVM	0.9058	0.9015	-0.5% (acceptable)	—

Key Advantage

- ✓ Language-agnostic
- ✓ Single hyperparameter (θ)
- ✓ Applicable before labelling
- ✓ **87.59% vocabulary reduction**

08 — KEY FINDINGS & TRADE-OFFS



Static Strategies — Modest Gains

TF-IDF already implicitly down-weights stopwords. NLTK removal adds only 0.82% vocab reduction. Best static: Extended + SVM at F1 = 0.9070.



NB Highly Sensitive to Noise

20.3% F1 gain confirms generative models suffer most from retaining uninformative tokens. Aggressive removal is essential for NB.



SVM Inherently Robust

Only 0.55 pp spread across all 5 strategies. LinearSVC's margin maximisation makes explicit removal largely redundant.



Adaptive = Best Overall Balance

87.59% compression, best NB/LR accuracy, 3.8× LR speedup — at a cost of only 0.5% for SVM.



87% of Vocab is Redundant

Most Reuters tokens have near-zero TF-IDF across all 57 categories — they carry no discriminative signal.



Production Recommendation

Use adaptive for NB/LR (better accuracy + faster). For SVM, strategy barely matters — choose none or extended.

CONCLUSION

Best Static

Extended + SVM

F1 = 0.9070, Accuracy = 0.9123

Best Adaptive NB

Adaptive + NB

F1 = 0.7160 (+20.3%)

Most Efficient

Adaptive + LR

73.9% faster, -1.4 pp F1

Max Compression

87.59% reduction

14,794 → 1,836 features

The adaptive TF-IDF strategy proves that corpus-specific, unsupervised vocabulary pruning substantially reduces computational cost while preserving or improving accuracy.

★ GitHub: <https://github.com/RohanBabhale1/Stopword-impact-analysis>

Thank You!